

# Transcriptomic profiling of Schlemm's canal cells reveals a lymphatic-biased identity and three major cell states

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## Abstract

Schlemm's canal (SC) is central in intraocular pressure regulation but requires much characterization. It has distinct inner and outer walls, each composed of Schlemm's canal endothelial cells (SECs) with different morphologies and functions. Recent transcriptomic studies of the anterior segment added important knowledge, but were limited in power by SEC numbers or did not focus on SC. To gain a more comprehensive understanding of SC biology, we performed bulk RNA sequencing on C57BL/6J SC, blood vessel, and lymphatic endothelial cells from limbal tissue (~4500 SECs). We also analyzed mouse limbal tissues by single-cell and single-nucleus RNA sequencing (C57BL/6J and 129/Sj strains), successfully sequencing 903 individual SECs. Together, these datasets confirm that SC has molecular characteristics of both blood and lymphatic endothelia with a lymphatic phenotype predominating. SECs are enriched in pathways that regulate cell-cell junction formation pointing to the importance of junctions in determining SC fluid permeability. Importantly, and for the first time, our analyses characterize 3 molecular classes of SECs, molecularly distinguishing inner wall from outer wall SECs and discovering two inner wall cell states that likely result from local environmental differences. Further, and based on ligand and receptor expression patterns, we document key interactions between SECs and cells of the adjacent trabecular meshwork (TM) drainage tissue. Also, we present cell type expression for a collection of human glaucoma genes. These data provide a new molecular foundation that will enable the functional dissection of key homeostatic processes mediated by SECs as well as the development of new glaucoma therapeutics.

## Significance statement

Schlemm's canal is a critical ocular drainage structure that maintains intraocular pressure with important consequences for glaucoma. Despite this, Schlemm's canal has been relatively

understudied with the molecular process that mediate its function, maintain its health, or that are compromised in glaucoma needing much deeper characterization. This study is important as it provides the most comprehensive molecular characterization of Schlemm's canal cells to date providing a wealth of molecular details. In addition to highlighting important biological pathways, this information will guide studies to develop new treatments against glaucoma.

### eLife assessment

This **valuable** study has characterized the unique expression of Schlemm's canal endothelial cells (SECs) using FACS-sorted specific cell bulk RNA-Seq and scRNA-/snRNA-Seq of mouse SECs. The **compelling** study identified novel biomarkers for SECs and molecular markers for two inner wall SEC states and outwall SECs in mouse eyes. Significant gene networks and pathways were elucidated for their potential contribution to glaucoma pathogenesis, providing targets for further research in relation to glaucoma.

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## Main text Introduction

Glaucoma is a leading cause of blindness affecting 80 million people (1 [link](#)). Elevation of intraocular pressure (IOP) is a major risk factor for glaucoma. Current glaucoma treatments reduce IOP using drugs and surgeries that decrease aqueous humor (AQH) formation or increase aqueous humor drainage (outflow) from the eye (2 [link](#)). Abnormally increased outflow resistance results in elevation of IOP and contributes to glaucoma. Schlemm's canal (SC) and the trabecular meshwork (TM), key tissues of the conventional outflow pathway, are critical in regulating IOP and ocular fluid homeostasis. The flow resistance of the conventional outflow pathway is an important determinant of IOP. SC is an endothelial vessel circumscribing the eye within the iridocorneal angle at the limbus, where the iris and cornea meet. The canal has inner and outer walls composed of SECs (3 [link](#)). The inner wall (IW) of SC is the final barrier that AQH must cross before it passes into the lumen of SC, which is connected to venous circulation by vascular branches known as collector channels. The IW is specialized for its drainage functions with higher expression of lymphatic genes and differing cellular morphology compared to the outer wall (OW) (4 [link](#)). The IW regulates aqueous humor drainage into the canal's lumen. IW cells respond biomechanically to regulate IW fluid permeability and drainage. Specifically, they respond to changes in ocular pressure and shear stress, generated by AQH flow, forming specialized drainage structures called giant vacuoles as well as pores through which AQH passes (5 [link](#)–7 [link](#)). However, much remains to be discovered about the mechanisms that determine outflow resistance and mediate outflow under both normal and pathological conditions.

Delineating the molecular control of outflow resistance and of inner wall fluid permeability is important for understanding IOP homeostasis and developing novel glaucoma treatments. A key site of resistance to outflow is located where SC and the adjacent juxtacanalicular meshwork (JCT) – a subpart of the TM – meet. The SC and TM are intimately linked at the cellular, biomechanical and functional levels (8 [link](#), 9 [link](#)). Both physical and molecular interactions between SC and TM cells participate in tissue formation, tissue maintenance and the regulation of resistance to outflow. The mechanisms by which the SC and TM coordinately regulate resistance to outflow and IOP are still being elucidated.

SC is a highly specialized vessel with its endothelial cells having properties of both blood vessels and lymphatic endothelial cells (BECs and LECs) (4). It develops by a process known as canalogenesis, which progresses through stages with features of angiogenesis, vasculogenesis, and lymphangiogenesis (4, 10–12). Analyses of the genes linked to glaucoma most strongly identify lymphangiogenesis and endothelial processes, strongly implicating dysfunction of SC (13). Despite this, SC is less studied than the TM and so we focus on SC in this study.

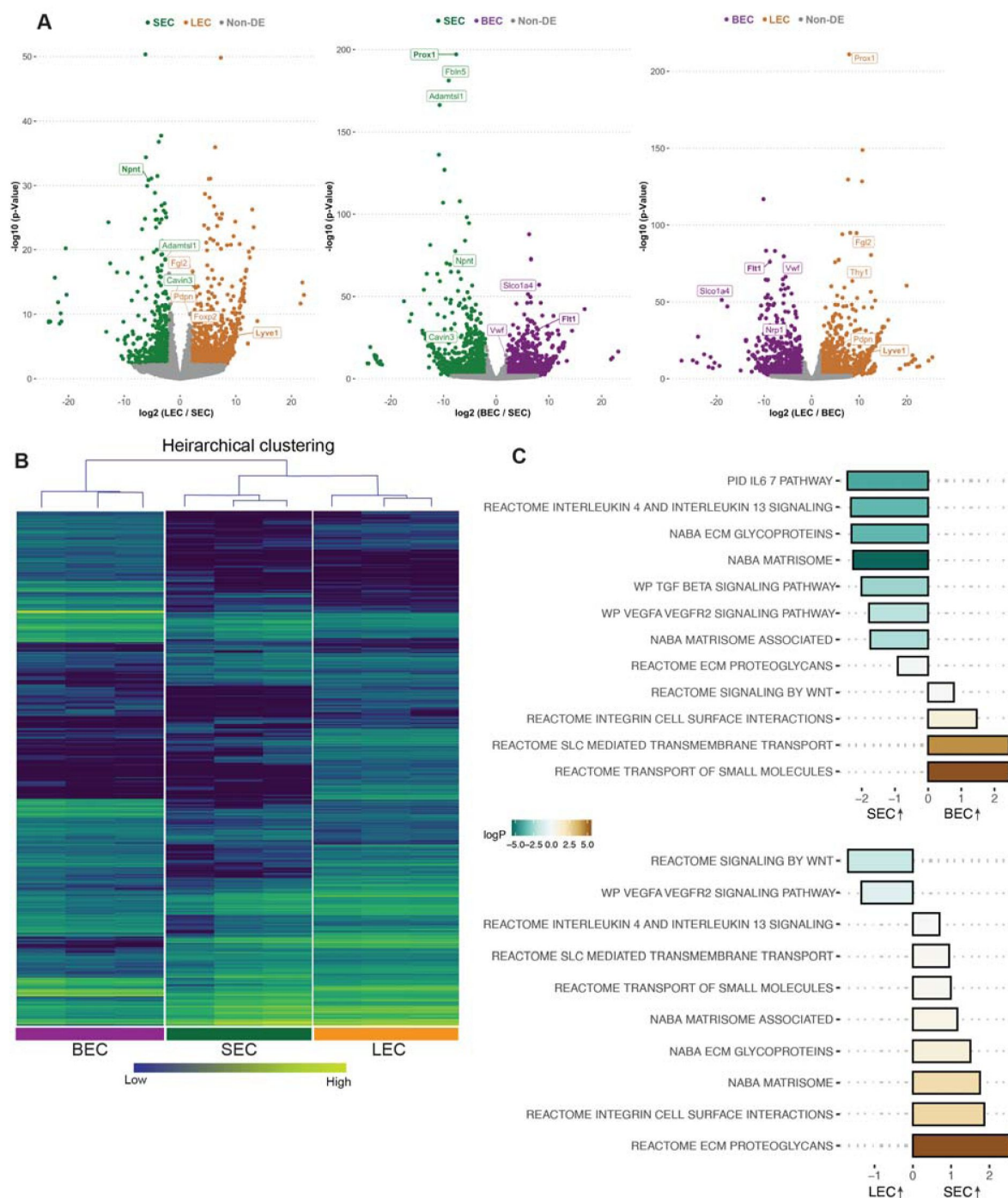
Single-cell RNA sequencing (scRNA-seq) is revealing the cellular heterogeneity and transcriptomic profiles of various ocular cell types, including the anterior segment from several species (14–18). These important studies included SECs, but were underpowered and limited for SC due to the low representation of SECs in the studied tissues and the very small numbers of SECs successfully sequenced. To more deeply and rigorously characterize SECs, we here provide a multi-modal transcriptomic analysis. We present bulk sequencing data for SECs, LECs and BECs from the same tissue sample, providing averaged data on SEC gene expression at depth for ~4500 SECs. We compare these bulk data to single cell (sc) and single nucleus (sn) RNA sequencing and then use the individual SEC transcriptome data to characterize SEC heterogeneity (across transcriptomes from 903 cells/nuclei from 2 strains of mice; C57BL/6J scRNA-seq 166, C57BL/6J snRNA-seq 375, 129/Sj scRNA-seq 362). Both the bulk and single cell resolution data confirm that SEC transcriptomes have similarities to BECs and LECs, but with an important bias towards lymphatic similarity. By sequencing a larger number of limbal SECs than previously reported, our study adds significant power and robustness, uncovering important new molecular characteristics of SECs. Building on the previous studies, it provides a more solid foundation to guide future work. Importantly, we identify unique gene expression features that distinguish inner and outer wall SECs, and we validate expression patterns using immunofluorescence (IF). Using pathway analysis and predicted ligand-receptor analysis, we provide critical new information about genes and pathways functioning in SC and about molecular interactions between SC and TM that are predicted to be central for tissue homeostasis, IOP control, and glaucoma. These unbiased methods greatly expand our molecular knowledge of SC.

## Results

### Schlemm's canal ECs are more similar to lymphatic than blood vessel ECs

To provide a more detailed understanding of the molecular nature of SECs, we compared the transcriptome of isolated limbal endothelial cells (SECs, BECs and LECs) obtained by bulk RNA sequencing. We developed a cell sorting method to isolate cells while preserving SECs. Limbal strips from Prox1-GFP mice were gently dissociated and dispersed cells were prelabeled with antibodies against LYVE1 and endomucin (EMCN). FACS was used to separate the dispersed cells into SECs (GFP+, EMCN+), LECs (GFP+, LYVE1+, EMCN-) and BECs (EMCN+ only). When we isolated mouse SECs with standard tissue dissociation procedures very few cells were obtained (<100 from 10 eyes), whereas our optimized protocol allowed isolation of sufficient cells for meaningful studies (1500-1700 SECs, 200-300 LECs, and 2000-2500 BECs from 10 eyes) (Figure S1).

Pair-wise comparisons of the bulk RNA-seq data for each purified cell type demonstrate that the transcriptome of SECs displays more genes in common with LECs than BECs (Figure 1). As LECs and SECs have the most similar transcriptomes, volcano plots of differentially expressed genes (DEGs) between LECs and SECs showed fewer DEGs than for BECs versus SECs or for LECs versus BECs (Figure 1A). *Npnt* was highly expressed in SECs, but absent in LECs and BECs. As expected, (4, 10) *Prox1* was the top DEG comparing SECs versus BECs and also LECs versus BECs (Figure 1A). The dendrogram distances after hierarchical clustering and the signature gene heat maps of the bulk RNA sequencing data further demonstrate that the transcriptome of SECs is closer to that of LECs than BECs (Figure 1B).



**Figure 1**

### SEC bulk transcriptome is more similar to LEC than BEC transcriptomes.

A. Pairwise comparison of bulk RNA sequencing data showing differentially expressed genes (DEGs) between cell groups, note larger number of non-DEGs in gray in plot of LEC vs. SEC compared to others. B. Hierarchical clustering of BECs, SECs, and LECs. C. Pathways enriched in SECs vs BECs (top panel), LECs vs. SECs (bottom panel) by GSEA analysis. BEC: Blood endothelial cell, LEC: Lymphatic endothelial cell, SEC: Schlemm's canal endothelial cell.

We next determined which canonical pathways were different between the cells based on their transcriptomes. Gene set enrichment analysis (GSEA) of SECs compared to BECs or LECs showed a common theme with extracellular matrix interactions (ECM glycoprotein, integrin cell surface interactions) being highly enriched in SECs compared to both other cell types (**Figure 1C**). TGF $\beta$  (19, 20) cell signaling pathway was also enriched in SECs when compared to BECs but not LECs. Additionally, interleukin pathways were enriched in SECs compared to both BECs and LECs indicating their specific potential to interact with immune cells (**Figure 1C**, Table 1, Table 2).

## Cell types in mouse limbal tissue

Using scRNA-seq, we successfully sequenced 9272 single cells from dissected limbal tissue (enriched for drainage structures) of C57BL/6J mice (**Figure S1**). An unbiased low-resolution clustering of single cells projected onto a Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) space identified 9 major clusters representing various anterior segment cell types (**Figure 2A**). Clusters were assigned to their cell types using previously characterized marker genes (14) as indicated in the figures (**Figure S2**). We then focused on the *Egfl7*-expressing EC (a commonly used endothelial marker, 469 cells) cluster for further sub-clustering and analyses (**Figure 2B**).

## SEC transcriptome is unique but closer to LEC than BEC

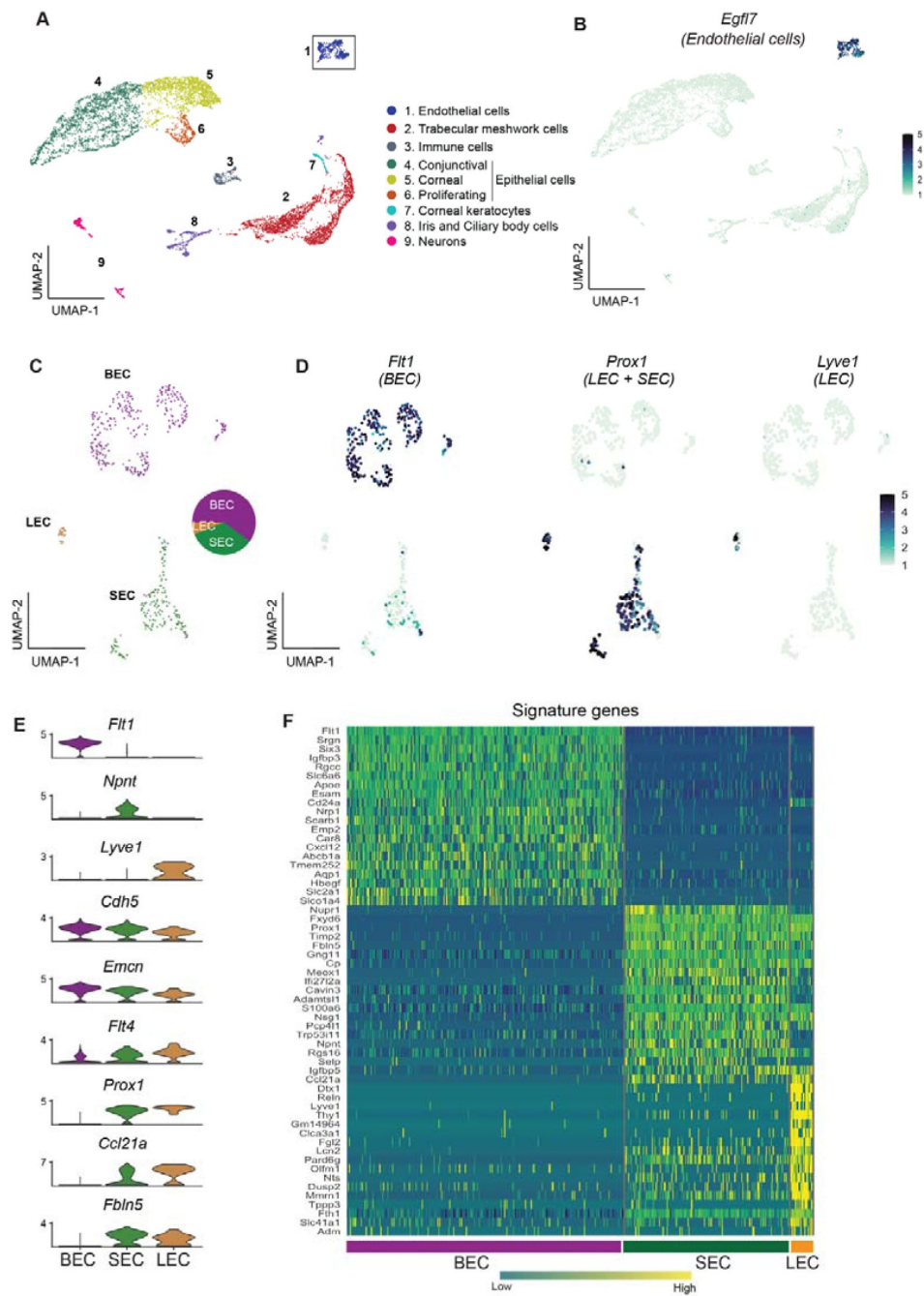
Using the strain C57BL/6J scRNA-seq data, the *Egfl7*-expressing ECs subclustered (unbiased methods) into three distinct sub-clusters (**Figure 2C**, D). Based on well characterized marker gene expression (4, 10, 14, 21) these sub-clusters were assigned as BEC (281 cells, high level of *Flt1* / *Vegfr1*), LEC (22 cells, high level of *Prox1* and *Lyve1*) or SEC (166 cells expressing *Prox1* but not *Lyve1*) (**Figure 2D**). Among ECs, *Npnt* is uniquely expressed in SECs, *Cdh5* and *Emcn* are expressed at high levels in BECs, but only at mid and low levels in SECs and LECs, respectively. *Flt4* (*Vegfr3*) is expressed at high levels in LECs, and at mid or low levels in SECs and BECs, respectively. *Prox1*, *Ccl21a*, and *Fbln5* are expressed in both LECs and SECs. *Prox1* and *Ccl21a* are expressed at higher levels in LECs but are not unique to either LECs or SECs (**Figure 2E**). A heatmap of the highly represented genes for each cell type indicates that the transcriptome of SECs more closely aligns with that of LECs than BECs (**Figure 2F**). The scRNA-seq data correlated well with the bulk sequencing data (**Figure S3**). When analyzing scRNA-seq data using Seurat, clusters are determined by calculating shared nearest neighbors followed by modularity optimization using a Louvian algorithm. To determine if the signature genes identified in these clusters identified by Seurat were robust and able to distinguish cell types in bulk RNA-seq samples, we used this signature gene list for unsupervised hierarchical clustering of the bulk RNA-seq data. This correctly identified the 3 cell types producing accurate SEC, BEC and LEC clusters. This demonstrates the robustness of the signature genes across independent samples using distinct statistical calculations (**Figure S3**).

## RNAseq enables detection of OW and IW transcriptomes

Previous studies reported higher representation of ECs when using snRNA-seq compared to scRNA-seq (22, 23) (and personal communication, Seth Blackshaw). Thus, we reasoned that snRNA-seq would capture more ECs and provide more power to discriminate SEC subtypes using unbiased methods. Using snRNA-seq, we analyzed 10,764 C57BL/6J nuclei (**Figure 3A**) and captured 1287 EC nuclei (11.9% of total nuclei). This is a clear enrichment over scRNA-seq (EC 5.1% of cells).

The scRNA-seq and snRNA-seq datasets integrated well and agreed with each other in all assessed ways (**Figure 3B**). Although snRNA-seq captured fewer distinct genes (median gene count per cell = 759) than scRNA-seq (median gene count per cell = 2603), endothelial cells from the two datasets had a high degree of correlation of gene expression (**Figure S3**).

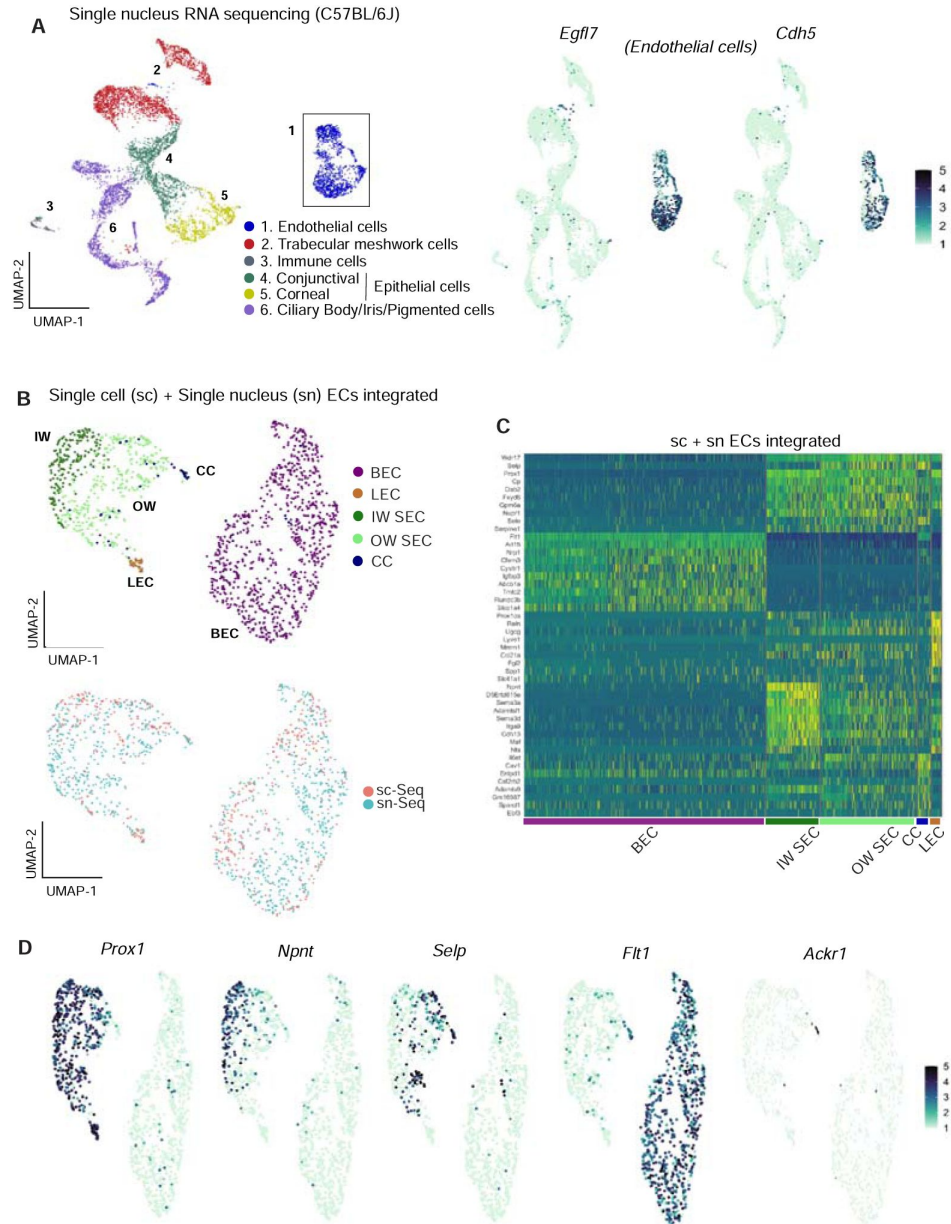




**Figure 2.**

### scRNA-seq data reveals robust SEC signature genes.

A. Clusters of cells from the limbal tissue represented on a UMAP space B. Expression of *Egfl7* in endothelial cells. C. Endothelial cell cluster is sub-clustered into BEC, LEC, and SEC. D. Expression of *Flt1* enriched in BEC, *Prox1* in LEC and SEC, and *Lyve1* in LEC identifying the three cell clusters. E. Violin plot of genes expressed BEC, SEC, and LEC from scRNA-seq data. F. Heatmap of signature genes from BEC, SEC, and LEC from scRNA-seq data. BEC: Blood endothelial cell, LEC: Lymphatic endothelial cell, SEC: Schlemm's canal endothelial cell.



**Figure 3**

### Integrating scRNA and snRNA-seq enables detection of outer and inner wall transcriptomes.

A. snRNA-seq of C57BL/6J limbal tissue identifies similar cell types as the sc RNA-seq but captures more endothelial cells (left panel). Expression of *Egfl7* and *Cdh5* in snRNA-seq endothelial cells (right panel). B. Integration of sc and snRNA-seq endothelial cells followed by sub-clustering identifies BECs, LECs, IW SECs, OW SECs, and CC. Integration of sc- and sn- RNA sequencing shows distribution across clusters (bottom panel). C. Heatmap of differentially expressed genes of the identified sub-clusters. D. SECs and LEC sub-clusters identified in B, expressing *Prox1*. In the SEC cluster, IW cells express *Npnt* and OW cells *Selp*, CCs express *Ackr1* and BECs robustly express *Flt1*. IW: Inner wall, OW: Outer wall, CC: Collector channels, BEC: Blood endothelial cell, LEC: Lymphatic endothelial cell, SEC: Schlemm's canal endothelial cell.

Integration of *Egfl7*<sup>+</sup> *Cdh5*<sup>+</sup> endothelial cells from the two datasets provided transcriptome data for 541 C57BL/6J SECs (166 scRNA-seq + 375 snRNA-seq).

Unbiased clustering of this larger number of SEC transcriptomes separated SECs into two distinct sub-clusters (**Figure 3C and D**). One cluster had higher expression of *Npnt* (representing inner wall, IW, cells; validation in results below) while the other had higher levels of *Selp* and *Sele* (representing outer wall, OW, cells). A small number of the OW cluster cells also had a limited, but specific, expression of *Ackr1* (alias *Darc*), which was previously suggested (14, 24) as a marker for collector channels endothelial cells. We manually separated these cells from the outer wall cluster and here forward denote them as collector channel (CC) cells. *Ackr1* is validated to represent CC's (see results below). Given the deeper sampling of transcripts of distinct genes in the scRNA-seq data, we manually separated our scRNA-seq SECs into IW, OW, and CC clusters using markers derived from the unbiased clustering of integrated data, (Figure S4).

## Additional scRNA-seq transcriptomes allows detection of two distinct IW cell states

We reasoned that adding more transcriptomes with deeper coverage (transcripts from greater number of distinct genes detected) would enable better resolution of SEC subclasses. To examine more transcriptomes and with deeper coverage than snRNA-seq can achieve, and to validate our findings in a genetically divergent strain of mice, we performed scRNA-seq on limbal tissue from 129/Sj mice. We successfully sequenced 8812 limbal cells of 129/Sj mice (median gene count per cell = 1144). All major cell types were represented in the 129/Sj limbal data (**Figure 4A**) and it compared well to the C57BL/6J data. We integrated the *Egfl7*<sup>+</sup> *Cdh5*<sup>+</sup> endothelial cells from 129/Sj and C57BL/6J datasets, a total of 1492 ECs (528 SECs) (**Figure 4B**). No statistically significant pathway differences were detected between the SECs of these strains, and a full analysis of the effects of strain background is beyond the scope of the current study (Table 3, shows a DEG list). Unbiased clustering of this integrated data detected three SEC states by further resolving the previously detected IW state into two subclasses that we call IW1 and IW2. IW1 cells are typically *Npnt*<sup>high</sup> and IW2 cells are often *Ccl21a*<sup>high</sup> while OW cells are often *Selp*<sup>high</sup>, but combinations of markers provide the best cell type resolution (**Figure 4E**, Table 5). We again identified and manually clustered a small group of *Selp*<sup>high</sup> cells that were *Ackr1*<sup>+</sup> CC cells (**Figure 4C** – E).

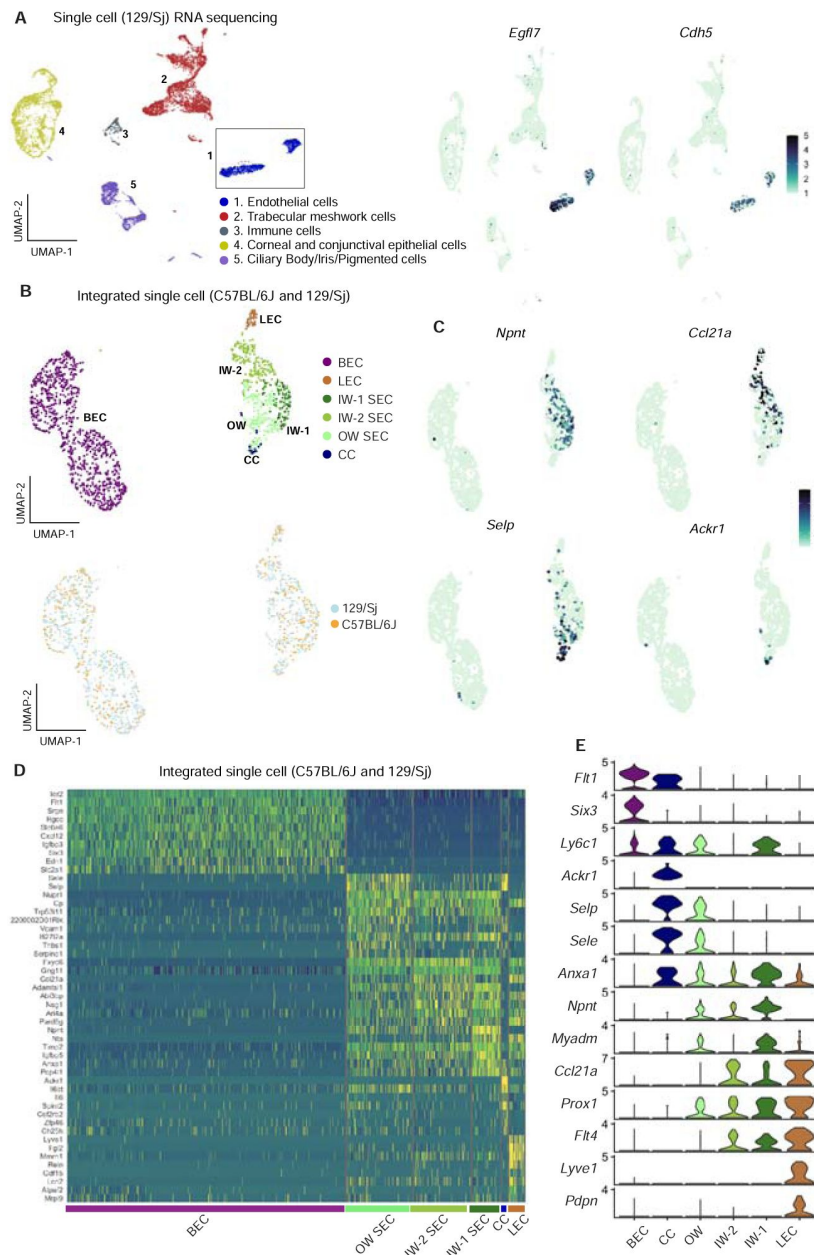
## Immunofluorescence assays validate transcriptomic resolution of OW and IW SECs

To determine the location of cells with detectable NPNT, CCL21A, and SELP proteins in eyes we performed IF on ocular sections and flat mounts of the anterior segment. NPNT was uniquely expressed at a high level in the extracellular matrix surrounding the IW of SC (**Figure 5A**) but was not detected in the OW of SC. NPNT was often detected to have a gradient of expression within SC, with detection at highest levels in the anterior portion of SC (and sometimes exclusively anterior detection). Using *in-situ* hybridization, we confirm the expression of *Npnt* in IW SECs with an anterior bias (Figure S5). CCL21A often had the reverse gradient, with the greatest (and sometimes exclusive) detection in the posterior portion of SC (**Figure 5A**, B; Figure S5). This suggests that the expression of NPNT and CCL21A may be modulated by local environmental differences such as aqueous humor flow rates.

*Ccl21a* has been previously used as an SEC marker in scRNA analyses of mouse data

(14). However, our current data and a previous study (11) clearly show that CCL21A is not unique to SC in the limbus being also expressed in limbal lymphatics (**Figure 5B**).

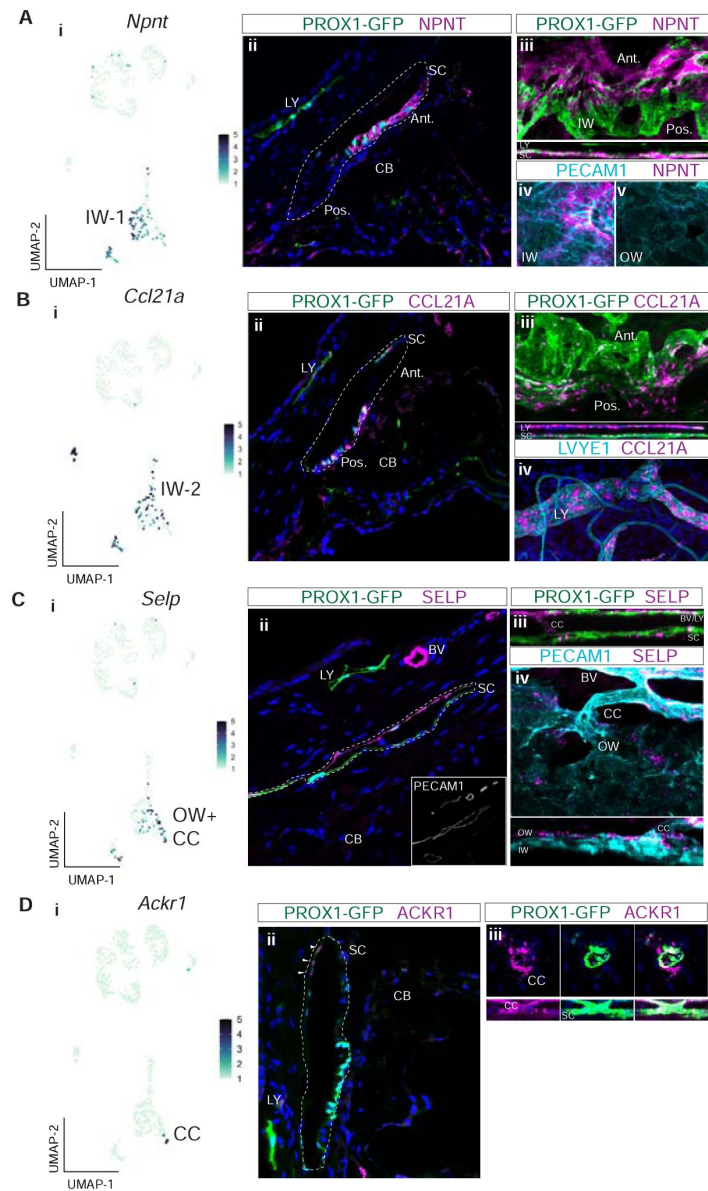




**Figure 4.**

### Integration of C57BL/6J and 129/Sj endothelial cells identifies IW states.

A. scRNA-seq of 129/Sj anterior segment tissue identifies various cell types similar to that in C57BL/6J single-cell RNA sequencing (left panel). Expression of *Egfl7* and *Cdh5* in snRNA-seq endothelial cells (right panel). B. Integration of B6 and 129/Sj endothelial cells followed by sub-clustering identifies BECs, LECs, IW1 SECs, IW2 SECs, OW SECs, and CCs (top panel). Integration of B6 and 129/Sj endothelial cells distributed across clusters (bottom panel). C. Sub-clustering identifies complementary expression patterns of *Npnt* and *Ccl21a* in IW1 and IW2 SECs, *Ackr1* in CCs, and *Selp* in OW SECs. D. Heatmap of differentially expressed genes of the identified sub-clusters. E. Violin plot showing differences in expression levels of various genes which as a combination defines individual sub-clusters. IW: Inner wall, OW: Outer wall, CC: Collector channels, BEC: Blood endothelial cell, LEC: Lymphatic endothelial cell, SEC: Schlemm's Canal endothelial cell.



**Figure 5.**

### Immunofluorescence validates cell types and discovers bias for discrete localization of IW1 and IW2 cells.

A. *Npnt* expression in a subgroup of SEC in scRNA-seq data (i) and corresponding immunofluorescence (IF) reveals high level of expression of NPNT in anterior portion of IW of SC in a frozen section (ii) and whole mount (iii and iv). B. *Ccl21a* is expressed in SECs and LECs (i) and corresponding IF reveals high expression in posterior portion of IW of SC in a frozen section (ii) and whole mount (iii and iv). C. *Selp* is expressed in OW SECs and CCs, a subgroup of SECs in single-cell (i) and corresponding IF (ii frozen section, iii-iv whole mount). D. *Ackr1* expression in a subset of CC cells (i) and corresponding IF (ii frozen section, iii whole mount). DAPI in blue labels nuclei in all panels. IW: Inner wall, OW: Outer wall, CC: Collector channels, CB: Ciliary body, LY: Lymphatic vessels, BV: Blood vessels SC: Schlemm's canal. Ant.: Anterior SC, Post.: Posterior SC.

SELP was detected in the outer wall of SC and some, but not all, BECs. This agrees with our transcriptomic data, where *Selp* was expressed in a significant subset of SECs by snRNA-seq, but minimally so in BECs. Whole mount staining of the anterior segment revealed SELP expression in OW SECs and ECs of CCs (**Figure 5C**). Taken together, the subgroup of SECs high in *Selp* expression must be OW SECs and CCs, while the subgroup of SECs with obvious *Npnt* expression are IW SECs.

Immunofluorescence detected ACKR1 in CCs and possibly very few OW cells (only where the CCs and OW meet, **Figure 5D**). We denote these cells as CC cells in this paper. The exact starting location of CCs is hard to discern, as they blend into the OW, and there are very few of these *Ackr1*-expressing cells in the transcriptome data. The best ACKR1 antibody that we have tested is FAB6695 which has been used to detect ACKR1 in endothelial cells (14). However, it binds non-specifically to another protein(s) in macrophages (25, 26). At the RNA level, the CC cells also express low levels of the SEC marker *Prox1* and have intermediate to high expression of the vascular markers *Flt1* and *Aqp1* (**Figure 2D**, 3D, 4E). Thus, we also assessed FLT1 by IF. Agreeing with both the transcriptomic data and our cell type designations, FLT1 was detected strongly in CCs and BECs, and to a weaker extent in a subset of SECs (Figure S6). Therefore, manual clustering of *Ackr1*<sup>+</sup> *Selp*<sup>+</sup> cells in the SEC cluster of both scRNA-seq and snRNA-seq accurately identifies CC cells (**Figure 3B**, 3C, and 4A - C).

## Other cell types within the major EC cluster and confirmation of greater lymphatic polarization of IW

Unless otherwise mentioned, further analyses were performed using the C57BL/6J scRNA-seq data because it had the deepest transcript coverage. In EC sub-clusters, we identified clustering based on sex-specific differences (Figure S6). We identified a smaller ‘male’ cluster and a larger ‘female’ cluster in SECs, BECs, and LECs. The smaller male cluster size reflects our use of a third of the number of female eyes. Genes such as *Xist* (involved in the X-inactivation process) and *Ddx3y* (a Y-chromosome-linked gene) were specifically expressed in the female and male clusters respectively. *Lars2* and *Aes* were two genes associated with female and male clusters respectively, consistent with previous studies in other endothelial cell types (27). Whether there are functional differences resulting from sex-specific gene expression differences remains to be determined.

A close examination of the BEC cluster in the scRNA-seq dataset revealed a sub-cluster of cells that highly expressed markers such as *Des* (*Desmin*), *Pdgfrb*, and *Rgs5*, all known markers for pericytes (28, 29). In addition to wrapping around blood vessels, IF analysis showed that the *Des*-expressing pericytes also associated closely with the CCs (Figure S6). In addition to pericytes, we identified arteries expressing *Sox17*, capillaries expressing high levels of *Esm1*, and veins expressing *Adamtsl1* and *Car4* in the BEC cluster (21, 30–32) (Figure S7).

Using IF we show expression of lymphatic regulatory molecules including *Prox1* and *Flt4* in SECs, and specifically an enrichment of lymphatic molecules in the IW of Schlemm’s canal further confirming (4) a greater lymphatic polarization of IW cells (Figure S7).

## Cell type assignment of high IOP and glaucoma genes

We determined the cell type-specific expression of genes associated with elevated IOP and glaucoma by previous genome-wide association (GWAS) studies (33–38) and of developmental glaucoma genes identified by genetic studies (**Figure 6A-B**, Figure S8). The majority of genes were expressed in endothelial cells and TM cells. A disease score relevance analysis (scDRS) of GWAS genes in the entire limbal tissue indicated that a significant number of these genes were expressed predominantly in TM, endothelial cells, ciliary body, iris cells, and corneal epithelial cells (**Figure 6C**). Of the genes expressed in endothelial cells, we analyzed

specific expression in subtypes and found that most genes were expressed broadly among endothelial cell types, supporting the importance of vascular regulatory functions in IOP and glaucoma biology (Figure 6D [↗](#)).

## Predicted receptor-ligand analysis reveals signaling pairs between TM and SC

Predictive algorithms trained on known ligand-receptor interactions (39 [↗](#)) support active signaling between TM and SC cells. For instance, secreted ANGPT1 ligand expressed by TM cells is predicted to interact with TEK receptor expressed by SECs. ANGPT1/TEK is known to be a key signaling pathway in the developing and adult SC with GWAS linking it to ocular hypertension and glaucoma (37 [↗](#), 40 [↗](#)–42 [↗](#)) (Figure 7A [↗](#)). We also identified secreted VEGFA, VEGFB, and VEGFC from TM with interaction partners on SC cells (4 [↗](#), 11 [↗](#), 43 [↗](#)). In addition to secreted factors, novel contact-based interactions were also predicted, including contact-based signaling between Junctional adhesion molecule 2 (JAM2) in TM cells and ITGB1 on SC cells.

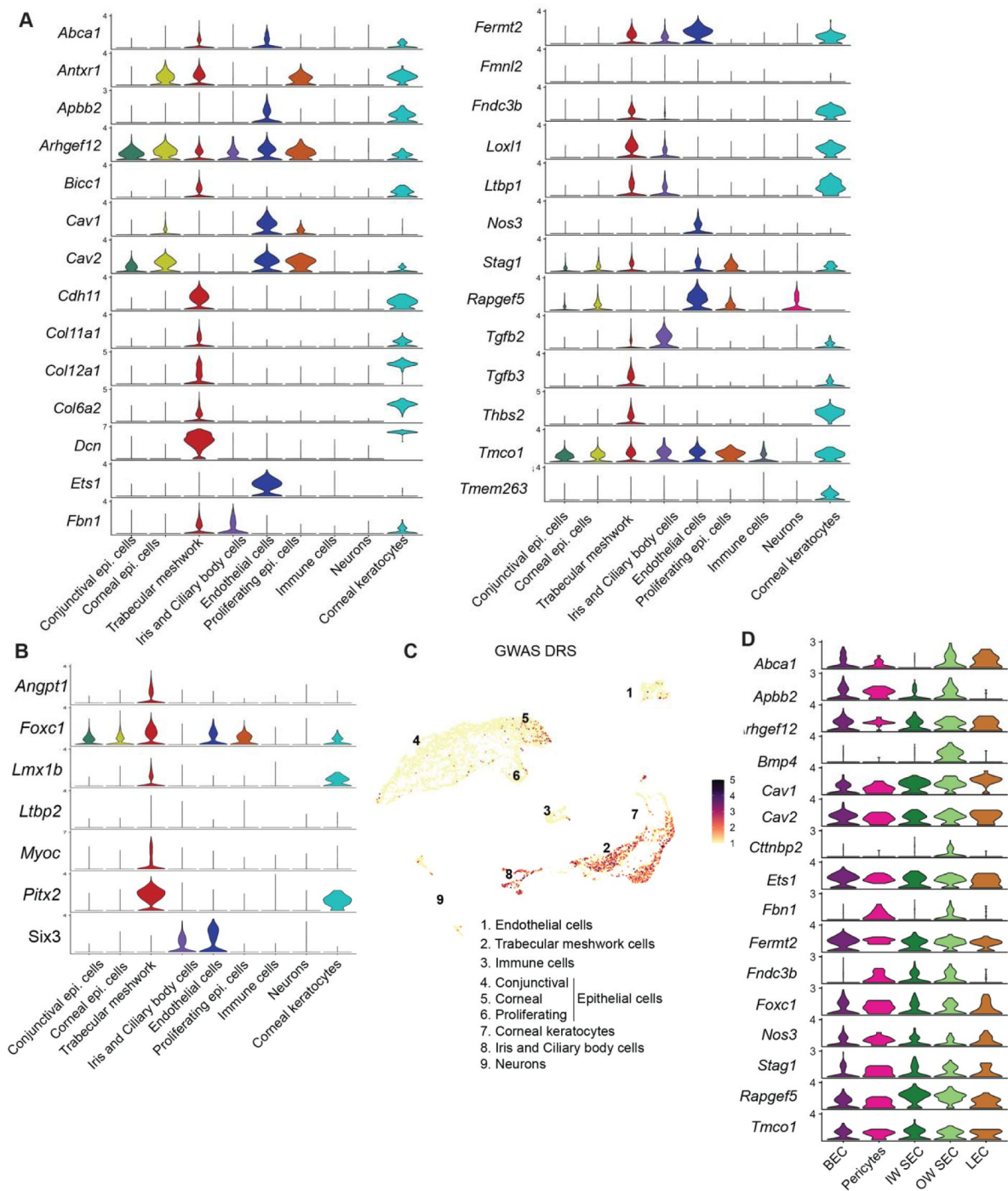
Conversely, we examined predicted interactions between ligands expressed by SECs with TM cells. Overall, fewer predicted interactions were found, arguing for a more important role for signaling between ligands originating from the trabecular meshwork in the development, maintenance, and function of SC than vice versa (Figure 7B [↗](#)).

## Gene ontology enrichment analysis highlights key cellular component/pathways in SECs

Gene ontology (GO) analysis of SC enriched genes was performed in comparison to a combination of the other endothelial cells and TM cells (Figure 7C [↗](#)). Enrichment of the cell-cell junction, membrane raft, and endocytic vesicles cellular components in SECs highlights their importance for SEC maintenance and cellular functions. Molecular pathways enriched in SECs were actin binding, cell adhesion molecule binding, SH3 domain binding, protein tyrosine kinase binding. Enriched biological processes in SECs were actin filament organization and regulation of actin filament-based processes, small GTPase-mediated signaling, cell substrate adhesion, and regulation of vascular development (Figure 7C [↗](#)).

These data highlight diverse pathways through which SC is likely to regulate outflow and hence IOP. For example, the nodal, membrane raft pathway includes the caveolin genes, *Cav1* and *Cav2*, which are functionally implicated in intraocular pressure modulation and glaucoma (44 [↗](#), 45 [↗](#)) as well as the caveolae-related genes, *Cavin2* and *Cavin3*, with roles in regulating the formation of caveolae (46 [↗](#)) (Figure 7D [↗](#)). Interestingly, *Il6st*, an immune component is also associated with this node and has recently been implicated in regulating the production and function of caveolins (47 [↗](#)). In the same node are other key vascular regulators that are expressed in SC such as *Kdr* (regulates various functions including permeability), *Tek* (endothelial barrier function), *Plvap* (fenestrae formation), and *Nos3* (vascular homeostasis) (Figure S9). GO analysis of IW or OW enriched genes compared to other endothelial cells was performed on the integrated C57BL/6J and 129/Sj scRNA-seq data. Among molecular pathways enriched in IW cells are receptor-ligand activity, cell adhesion molecule binding, integrin binding, and calcium-dependent protein binding. Among molecular pathways enriched in OW cells are growth factor binding and glycosaminoglycan binding indicating prominent roles in regulating immune function and chemokine activity in SECs (Figure S10).



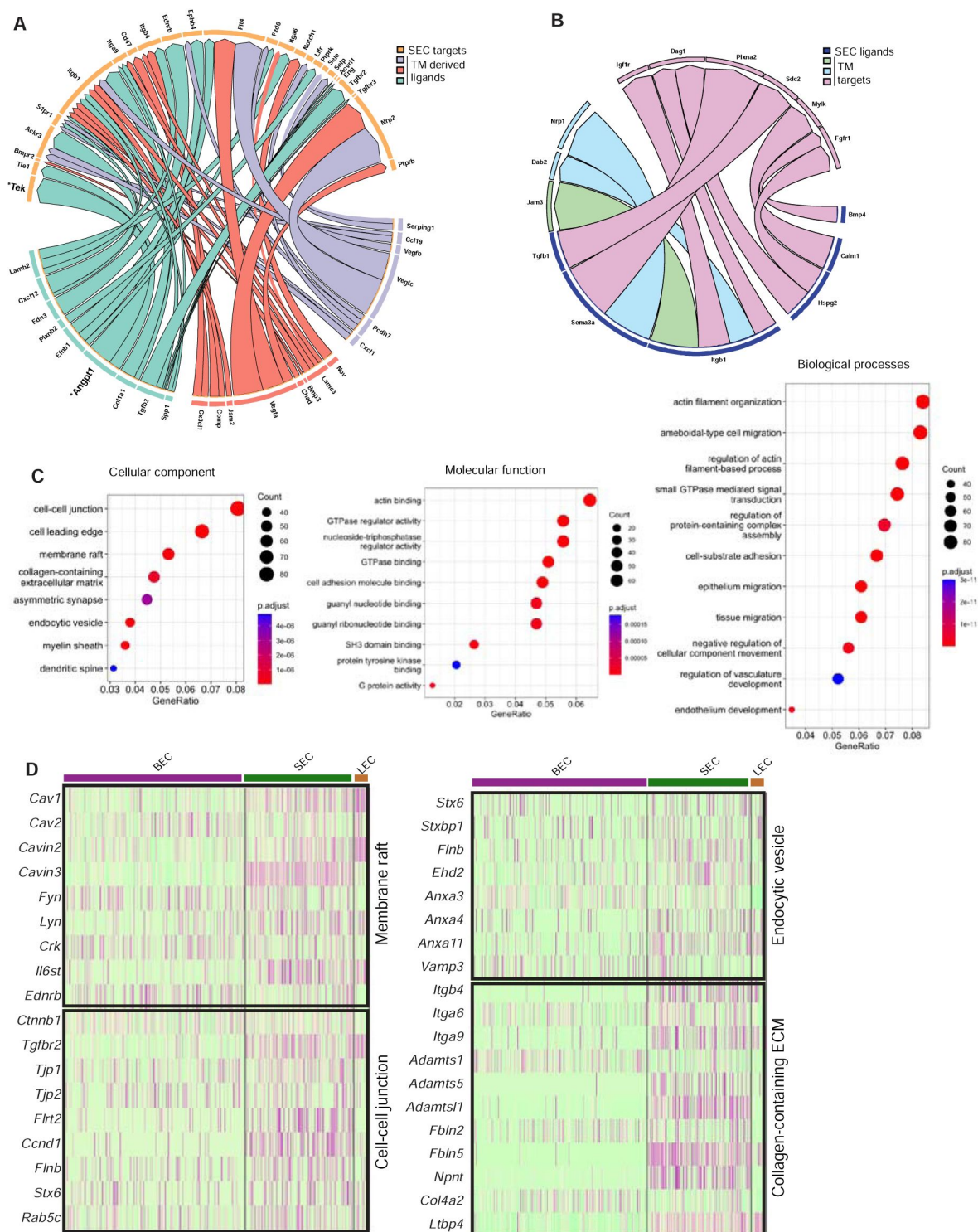


**Figure 6.**

### Cell-type assignment of high IOP and glaucoma genes.

A. Expression levels of gene list obtained from GWAS and genetic analyses associated with glaucoma and elevated IOP. B. Expression levels of genes involved in developmental forms of glaucoma. C. Disease relevance score (DRS) of genes from GWAS studies associated with glaucoma and elevated IOP. D. Expression levels of specific genes in subtypes of endothelial cells.





**Figure 7**

Predicted ligand-target analysis reveals signaling pairs between TM and SC. **A**, **B**. Circos plots depicting predicted ligand-target interaction between genes expressed in SECs and those in surrounding trabecular meshwork cells. **C**. Gene ontology analysis of genes enriched in SC. **D**. Heatmap of genes in individual enriched pathways showing their expression in endothelial cell subsets.

## Discussion

### A multi-modal approach to understanding Schlemm's canal

Single-cell resolution RNA-seq is revolutionizing molecular knowledge of cell types, cell states, and intercellular interactions in health and disease. Both scRNA-seq and snRNA-seq methods are greatly enabling. They overcome the previous necessity to purify individual cell types from complex tissues, and the associated technical issues with doing so. Importantly, they provide previously unavailable information about cell type and cell state heterogeneity. Nevertheless, tissue preparation methods still have to be optimized for specific tissues and cell types. Different preparation methods capture substantially distinct proportions of some cell types and/or different depths of transcriptomic data. Comprehensive analysis of desired cell types in a tissue may require the use of a combination of complementary, individually optimized preparation methods and sequencing approaches as presented here for SECs. We utilized bulk- RNA-seq of purified SECs from C57BL/6J mice as well as our current best optimized methods for scRNA-seq and snRNA-Seq to provide a robust analysis of the B6 limbus focusing on SEC transcriptomes. We successfully analyzed far greater numbers of SEC transcriptomes (541 total B6 transcriptomes, Table 4 summarizes all datasets) than previous studies ([14](#), [15](#), [17](#), [18](#)).

### Deep transcriptome data and lymphatic biased identity of SECs

Despite their central role in assessing cellular heterogeneity, both snRNA-seq and scRNA-seq currently sample a much smaller proportion of the transcriptome than the averaged transcriptome provided by bulk-RNA-seq of a purified cell type. Our bulk-RNA-seq data (Table 6) provide the deepest analysis of the SEC transcriptome to date identifying >9000 expressed genes (for 3 biological replicates representing approximately 1500 cells each) at FPKM > 5. Comparing this bulk SEC data to bulk data we generated for LECs and BECs (from the same limbal tissue samples) we observe a lymphatic biased identity for SECs. This differs from the conclusion in a recent scRNA-seq study ([18](#)), but is consistent with previous studies showing expression of lymphatic regulatory molecules including *Prox1* and *Flt4* in SECs ([4](#), [10](#), [11](#)) and enrichment of lymphatic molecules in the IW of SC ([4](#)). Adding to this, pseudo-bulk comparisons show that our scRNA-seq data agree well with our bulk data, again revealing the lymphatic biased expression pattern of SECs, and independently validating our conclusions based on bulk-seq. Our C57BL/6J scRNA-seq is of high quality with greater transcriptome representation (median ~2600 genes per cell) than is often achieved in limbal cells. Our re-analysis of the scRNA-seq data in the conflicting publication indicated that that study was underpowered (for example capturing ~5 LECs and ~90 SECs ([18](#))) and upon integration and analysis their SECs co-clustered with ours (not shown, we did not include their cells in following studies as they differed from ours in being both from an albino strain and treated with Y27362, a ROCK inhibitor. Overall, our findings significantly expand knowledge of the molecular relationship between SECs, LECs and BECs ([4](#), [8](#)) and they will support the field in formulating ideas and testing functions.

### snRNA-Seq allows identification of inner and outer wall SECs

snRNA-seq can have various advantages. It can allow the isolation of nuclei from frozen tissues. This theoretically helps to maintain transcriptomes as close to their natural state as possible and allows easy preservation of tissues for future analyses. Additionally, it can proportionally enrich for endothelial cells. For this reason, we used snRNA-seq in our studies.

ECs are very sensitive to preparation method and are routinely underrepresented using standard scRNA-seq methodologies, including in previous studies of the anterior segment and ocular drainage tissues ([14](#), [18](#)). On the other hand, snRNA-seq assays nuclear RNA, versus whole-cell RNA, and typically provides lower depth of transcriptome coverage than scRNA-seq. Our

snRNA-seq data (strain C57BL/6J) agree well with our bulk and scRNA-seq transcriptome data and again supports the lymphatic biased expression pattern of SECs. Our snRNA-seq data integrated well with the scRNA-seq data and provided proportionally far more SECs (Results).

The added power of increased SEC numbers allowed us to distinguish the transcriptomes of IW and OW SECs for the first time.

## Enhanced identification of different SEC states using further scRNA-seq

In addition to sampling greater number of cells, deeper sampling of the transcriptome of each cell leads to more robust pathway analyses and improved resolution of heterogeneous cell states. Currently, popular data integration methods anchor datasets based on the genes that are sampled. This means that extending data by adding more cells with low sampling is not as informative as adding more cells with more deeply sampled transcriptomes (adding data with lower sampling may even be detrimental, with the balance between the added statistical power of increasing cell number and depth of transcriptome sampling being important). Despite the general value of snRNA-seq, it generally achieves lower depth of transcriptome coverage, with greater technical variation in sampling of specific genes. This means that snRNA-seq data generally provide lower power for pathway enrichment analyses than scRNA-seq (as is true in our data).

Thus, to both provide further scRNA-seq data to enhance the resolution of SEC subtypes and to allow a cross-strain comparison, we next sequenced limbal tissue from strain 129/Sj mice (48 [48](#)). We obtained high quality data with standard coverage (362 SECs and median 1144 genes per cell, Table 4). The strain 129/Sj and B6 data integrate well and allowed stratification of SECs into 3 different states, including 2 IW states, for the first time. We named these classes OW, IW1, and IW2 based on marker studies and IF (see below). We refer to these SEC classes as states rather than subtypes because, despite their unbiased statistical separation, they have remarkably similar transcriptomes with no or very few detected pathway differences (at the currently achieved depth of sampling). Their transcriptomes may differ more due to the effects of local environmental differences (including exposure to different AQH flow parameters) than differing developmental trajectories. With greater sampling and deeper transcriptomic depth, it is likely that additional SEC cell states/types will be identified. Pathway enrichment analysis identified a few significantly different pathways between IW and OW cells. IW1 cells are enriched for the calcium-dependent protein binding and cell adhesion molecule binding pathways which makes intuitive sense, since SC contains many vesicles, suggesting that membrane fusion events must be important for normal function, likely in pore formation within drainage structures known as giant vacuoles. Membrane fusion in other cell types is regulated by calcium binding to various proteins including calmodulin (*Calml1*) (49 [49](#)). Inner wall cells have a discontinuous basement membrane and effective cell adhesion is necessary as the pressure gradient pushes them away from their basement membrane. Additionally, various cell adhesion molecule and related signaling can impact the stability of cell junctions, again with the possibility of modulating resistance to AQH drainage through the IW (Kizhatil et al., 2023 Biorxiv).

## Marker genes for OW and IW cells

Analysis of our scRNA-seq and IF data identified *Selp* as an OW marker among SECs. In agreement with this, Weibel-Palade bodies that produce, store, and release P-Selectin (SELP) are enriched in OW SECs (50 [50](#)). *Selp* was also expressed in CCs and some vascular endothelial cells. CCs also express *Ackr1* ((14 [14](#)) and Results) and have higher *Flt1* expression than OW and IW SECs. *Flt1* has been previously shown in SC (18 [18](#)), with robust expression based on IHC in one study (51 [51](#)). *Flt1* RNA is expressed at low levels and its protein is undetectable by IF in most SECs in our study in both the strain C57BL/6J and 129/Sj SECs. The reasons for this difference are not clear. Our integration and reanalysis of the scRNA-seq data from one study (18 [18](#)) showed that their SECs had similar *Flt1* expression patterns to ours (not shown).

Notably no markers have completely unique gene expression in any of these SEC classes and a combination of markers will provide best specificity for the majority of SECs (**Figure 4E**). Despite this, and due to differences in expression levels, IF looking for robust expression is a reasonable initial approach using some of the key marker genes for which reliable antibodies are available (e.g. NPNT and CCL21A). Along these lines, our results agree with previous studies (18) identifying *Npnt* as a marker for SECs among EC subtypes. *Npnt* promotes endothelial migration and tube formation (52) and thus could be key in the formation and maintenance of the adult SC. We extend previous understanding, by showing that *Npnt* is enriched in both IW1 and IW2 in comparison to OW cells. In fact, *Npnt* is most highly expressed in IW1 cells, and its transcript was detected in 89% of IW1 versus 26% of IW2 and 32% of OW cells (**Figure 4E**, Table 5). *Ccl21a* was also recently reported in SECs (11, 14). We extend this by showing that the *Ccl21a* gene is robustly expressed in IW2 cells but is typically expressed at low levels in IW1 and OW cells. Its transcript was detected in 45% of IW2 cells, 33% of IW1 and 21% of OW cells. Thus, high *Npnt* or high *Ccl21a* expression are key markers for these SEC classes (NPNT for IW1 and CCL21A for IW2). In agreement with our previous study demonstrating polarization of SC with a bias for robust expression of lymphatic molecules in the IW versus the OW (4). IW cells have higher expression of key lymphatic genes including *Flt4*, *Prox1* and *Ccl21a*. Assessing various marker genes by IF confirmed the identification of these OW and IW cell states. Additionally, an analysis of SECs in publicly available dataset (18), showed a similar distribution of *Npnt*, *Ccl21a*, and *Selp* expression in SECs (Figure S5) which further bolsters our findings.

## Potential regulators of AQH flow and immune cell transit expressed by IW SECs

In addition to validating the identification of OW and IW transcriptomes, our IF analyses further strengthened the biological separation of IW1 and IW2 SECs. IF revealed a bias for IW1 cells (marked by robust NPNT) to be located more anteriorly in SC than IW2 cells (marked by robust CCL21A). IW2 cells have a bias to be localized more posteriorly in SC. In mice, the anterior portion of the SC IW is likely to experience higher transmurial AQH flow rates based on the more open, anterior angle tissue anatomy, while the posterior portion of the IW is likely to experience lower flow rates. The posterior portion of SC has a differing local environmental context compared to the anterior portion being more closely opposed to: 1) the iris/uvea, 2) the more robust posterior trabecular meshwork and 3) the ciliary muscle and ciliary body. Together, this suggests that *Npnt* may respond to (and/or be a marker of) either local AQH flow rates, to local transport of ligands in AQH that are carried by flow, or to other local environmental differences. Since it is an extracellular matrix molecule (53), NPNT may even directly modulate the local regulation of flow. Further studies are needed to assess these possibilities and are already underway. Interestingly, the IW2 marker CCL21A is a potent chemoattractant ligand. It engages the CCR7 receptor on lymphocytes and antigen-presenting cells (APCs). *Ccl21a* is prominently expressed in lymphatics and lymph nodes and directs the migration of immune cells into lymphoid tissues (54, 55). Low fluid flow environments create a gradient of CCL21A that then directs cellular homing (54). We speculate that CCL21A may have similar functions in SC, which in addition to draining aqueous humor is intimately associated with immune cells, some of which transit the IW when exiting the eye (4, 56). SC is likely important in the transit of immune- tolerizing cells that are exposed to antigens in the anterior chamber, exit the eye and then travel to the spleen to establish a form of immune tolerance to the antigen captured in the eye, a process known as anterior chamber associated immune deviation (ACAID) (56).

## Macrophages, pericytes and collector channels

We previously reported a chaperoning role for macrophages in SC development, while adult SC is intimately associated with macrophages (4). It seems likely that macrophages associated with SC and/or present in the TM will have important roles in determining AQH outflow/IOP in health and disease. Our scRNA-seq data did not have adequate power for pathway analysis of macrophages, but further studies are underway. Our scRNA-seq also identified pericytes. IF



demonstrated that *Desmin*-expressing pericytes are present along the CCs. The physical association of pericytes with CCs suggests an important role in regulating outflow resistance at a site immediately distal to SC, but further experiments are needed to test this.

## Pathways in SECS

Gene ontology (GO) analysis of our scRNA-seq identified molecular pathways and hub genes that can be used to test hypotheses regarding SC function in regulating IOP. We identified genes involved in a variety of biological processes that are enriched in SECS. These processes include actin filament organization, cell substrate adhesion, regulation of vascular development, and tissue migration. Importantly, cell-cell junction pathway genes were enriched in SECS, including *Tjp1*, *Tjp2* and *Ctnnb1*. These genes have important roles in the formation and regulation of tight junction proteins in other systems (57 [↗](#)). Integrin genes involved in regulating collagen-containing extracellular matrix such as *Itga6*, *Itga9*, and *Itgb4*, fibulin genes with roles in maintaining the integrity of basement membrane such as *Fbln2* and *Fbln5*, and nephronectin (*Npnt*) an ECM molecule that regulates cell-cell adhesion (58 [↗](#)) are all expressed in SECS. Our GO analysis indicates enrichment of ‘membrane raft’ pathway genes in SECS including *Fyn*, *Lyn*, and *Crk* kinases in SECS. Recently we demonstrated a role for FYN, a Src family kinase in regulating AQH outflow and IOP (Kizhatil et al., 2023 Biorxiv). FYN is activated by IOP elevation and phosphorylates VECAD at specific tyrosine residues, thereby increasing the permeability of SEC cell junctions. Functionally analyzing hub genes identified in our pathway analyses is likely to improve our understanding of additional mechanisms that are active within SECS. Enhancing knowledge of pathway details in SECS, our bulk data provide much greater depth of transcript sampling and is valuable for extending the number of identified pathway members beyond the scRNA-seq data.

To provide important information on predicted ligand receptor signaling between the TM and SC, we used NicheNet (39 [↗](#)) to identify matching ligand receptor pairs present on SECS and TM cells. A list of key ligand receptor pairs is presented in **Figure 7** [↗](#). Supporting this approach and consistent with previous studies (4 [↗](#), 18 [↗](#), 40 [↗](#)), this analysis shows that *Angpt1* is produced by TM cells while its receptor TEK is highly expressed in SECS. Disturbances in this pathway are known to result in IOP elevation and developmental glaucoma while GWAS implicates it in POAG. Since IOP elevation is a key risk factor for glaucoma, we also provide a summary of the key cell types in which key genes implicated in IOP elevation and glaucoma though GWAS are expressed.

In summary, our study provides a deep characterization of SECS including the identification of 3 distinct cell states of great value to understanding both SC biology and IOP elevation in glaucoma. Although we focus on SC here, a wealth of data is provided for other anterior segment cell types that will be the focus of future studies.

## Methods

### Mouse strain, breeding, and husbandry

All animal procedures were performed according to the protocols approved by Columbia University’s Institutional Animal Care and Use Committee. Prox1-GFP BAC transgenic mice (Tg(Prox1-EGFP)KY221Gsat/Mmcd) were maintained on C57BL/6J (B6) background. B6 mice were obtained from JAX (Strain #:000664).

### Bulk RNA sequencing

The anterior segment of the eye including the limbus was dissected from each of ten 3-month- old Prox1-GFP mice to make 1 biological replicate of each cell type. The iris, cornea, and ciliary body were gently and carefully removed using forceps to prevent damage to the delicate aqueous



humor drainage structures. A strip of limbal tissue 1-2 mm wide consisting of the aqueous humor outflow tissues SC and TM, limbal blood vessels, and lymphatics was dissected from each anterior segment. Limbal strips were dissected in ice-cold PBS. Strips from 20 eyes were pooled and dissociated paying strict attention to osmolarity (290-300 mOsm) and mechanical force, using 1mg/ml Collagenase Type 4 (Worthington Biochemical Cat. no. LS004188) to preserve membrane proteins for the staining cell surface marker for flow cytometry. Single-cell suspension was stained with Lyve1 conjugated to eFluor 660 (ALY7 eBioscience Cat. no. 14-0443-82), Endomucin conjugated to phycoerythrin (eBio V.7.C7 Cat. no.14-5851-82), and propidium iodide to mark dead cells. Forward and side scatter gates were first used to eliminate events with low scatter which include debris, cell fragments and pyknotic cells. Then propidium iodide positive dead cells were gated out. Further gating on the viable cells was applied such that distinct population of cells were isolated a) SECs: GFP<sup>+</sup>Lyve1<sup>+</sup>, b) LECs: GFP<sup>+</sup> Lyve1<sup>+</sup>, c) GFP<sup>+</sup> BECs: Endomucin<sup>+</sup>. A total of 1500-1700 cells for SEC, 200-300 cells for LEC and 2000-2500 cells for BEC were collected. Three biological replicates were processed for each of SECs, BECs, and LECs. Raw data is available in GEO with accession number GSE272434.

## Processing of tissue for single-cell RNA sequencing

The anterior segment of the eye including the limbus was dissected from 3-month-old Prox1- GFP mice. A strip of limbal tissue 1-2 mm wide consisting of the aqueous humor outflow tissues SC and TM, ciliary body, iris, and cornea was dissected from each anterior segment in ice-cold Dulbecco's modified Eagle's medium DMEM. To enrich for aqueous humor outflow tissues, limbal strips from 6 C57BL/6J eyes (4 females, 2 males) were pooled for a single run of C57BL/6J scRNA-seq. C57BL/6J scRNA-seq was performed on two separate pools of eyes for a total of 12 eyes (8 females, 4 males). For 129/Sj scRNA-seq, 6 eyes (3 females, 3 males) were pooled for a single run.

Single cell dissociation was performed using Papain and Deoxyribonuclease I (Worthington Biochemical Cat. no. LK003153) for 20 min at 37°C and stopped using Earl's balanced salt solution (EBSS- Thermo Fisher Scientific, Cat no. 24010-043). Cells were triturated using an 18- gauge needle, centrifuged at 300g at 4 °C, washed with cold DMEM, and filtered using a 100 µm cell strainer. Cells were resuspended in cold DMEM and placed on ice immediately. Cells were counted using Countess II automated cell counter and processed for 10x library preparation immediately (see below).

## Processing of tissue for single nucleus RNA sequencing

The anterior segment of the eye including the limbus was dissected from 3-month-old Prox1- GFP mice. A strip of limbal tissue 1-2 mm wide consisting of the aqueous humor outflow tissues SC and TM, ciliary body, iris, and cornea was dissected from each anterior segment in ice-cold Dulbecco's modified Eagle's medium DMEM. To enrich for aqueous humor outflow tissues, limbal strips from 6 eyes (4 females, 2 males) were pooled for a single run of snRNA sequencing. Dissociation was performed using Papain and Deoxyribonuclease I (Worthington Biochemical Cat. no. LK003153) for 20 min at 37°C and stopped using Earl's balanced salt solution (EBSS). Cells were placed on ice and further nuclei lysis steps were performed according to 10X nuclei isolation from cell suspensions protocol (CG000124). Briefly, cells were lysed in lysis buffer for 45 seconds, resuspended in nuclei wash and resuspension buffer, and centrifuged for 10 minutes at 500rcf at 4 °C. Pellet was resuspended in nuclei wash and resuspension buffer and passed through a 40 µm cell strainer. An aliquot of nuclei suspension was stained with SYTOX green to count nuclei using Countess II automated cell counter.

## Single-cell and single-nucleus RNA sequencing

Single-cell and single-nucleus RNA-seq was performed at the single-cell sequencing core in Columbia Genome Center. Single cells or nuclei were loaded into chromium microfluidic chips with v3 chemistry and barcoded with a 10x chromium controller (10x Genomics). RNA from the

barcoded cells was reverse-transcribed, and sequencing libraries were constructed with a Chromium Single Cell v3 reagent kit (10x Genomics). Sequencing was performed on NovaSeq 6000 (Illumina). Raw data is available in GEO with accession number GSE271132.

## Analysis of sequencing data

**Bulk RNA sequencing:** Raw reads were aligned to mm10 reference genome by STAR and summarized to gene counts. After transcripts Per Kilobase Million (TPM) normalization, principal components analysis (PCA) was applied to assess technical replication and remove outlier samples. DESeq2 models were fitted to identify genes differentially expressed between blood vessel endothelial, SC endothelial, and lymphatic endothelial cells, where a threshold of log (fold change)  $> 2$  and  $FDR < 0.05$  was used to identify a total of 3,063 variable genes between three pair-wise comparisons. TPM of differentially expressed genes were shown as a heatmap between samples. The resulting log (fold change) values for each gene were plotted against those obtained from single-cell sequencing data described below.

**Single-cell and single-nucleus RNA sequencing:** Raw reads were mapped to the mm10 reference genome by 10x Genomics Cell Ranger pipeline. Seurat v3 ([59](#)–[62](#)) was used to conduct all single cell and single nucleus sequencing analyses. Briefly, the dataset was filtered to contain cells with at least 200 expressed genes and genes with expression in more than three cells. Cells were also filtered for mitochondrial gene expression ( $<20\%$  for single-cell and  $<5\%$  for single-nucleus, mitochondrial genes are rarely represented in snRNA-seq). The dataset was log-normalized and scaled. Unsupervised clustering was performed using a resolution parameter of 0.1, followed by manual annotation of Seurat clusters. Sub-clustering of individual clusters was also performed in an unsupervised manner with a resolution parameter of 0.5. Biological incompatibility based on gene expression was used to identify doublets. The two C57BL/6J single cell datasets were merged after QC. They were technical replicates, experimentally processed at the same time, and so batch effects resulting from sample preparation, sequencing technology, etc. do not apply. The merged C57BL/6J single cell dataset was integrated with the C57BL/6J single nucleus dataset post-QC, because of expected batch-effects between datasets arising from differences in sample preparation, sequencing technology, and alignment of reads (inclusion of intronic reads in snRNA-seq). Similarly, the merged C57BL/6J single cell dataset was integrated with the 129/Sj single cell dataset post-QC. Briefly, samples were log-normalized followed by the selection of highly variable genes and a reduction of dimensions using canonical correlation analysis. The unsupervised clustering with a resolution parameter of 0.1 for both single cell and single nucleus sequencing data was represented on a common UMAP space, and cluster identity was assigned based on the expression of various known genes. Marker genes were identified using Wilcoxon test implemented in Seurat using default parameters. Data can be visualized in the Broad Institute's Single Cell Portal with accession numbers SCP2293 (cell) and SCP2317 (nucleus).

**scDRS:** From genome-wide association studies of glaucoma ([33](#)–[38](#)) we curated a list of genes associated with the disease and applied single-cell disease relevance score ([63](#)) to assess glaucoma disease risk for each single cell, using default parameters.

**GO analysis:** Gene ontology and pathway enrichment analysis was performed by generating a list of genes enriched in individual endothelial cell groups by comparing against a background expression 'universe' of genes expressed in endothelial cells and trabecular meshwork cells (p value cut-off 0.01) ([64](#)) using R package ClusterProfiler ([65](#)).

**GSEA analysis:** Gene set enrichment analysis (GSEA) was performed with a pre-ranked list of genes from bulk RNA sequencing data with adjusted P-values  $<0.05$  and log fold change between groups  $>2$  were considered significantly different. GSEA was performed against canonical pathway consisting of 3090 gene sets.

Predicted ligand-receptor interactions: Predicted ligand-target links between interacting cells was performed using NicheNet (39). Briefly, expression of genes in cell types is linked to a database on signaling and gene regulatory networks curated based on prior information to make viable predictions on potential interactions between cell types.

## Immunofluorescence

Whole mounts of anterior segment of the eye were obtained from B6. Prox1-GFP mice or B6 mice and processed as described in (4). Briefly, mice were cervically dislocated and eyes harvested with the optic nerve intact and fixed in ice-cold 4% PFA for 2 hours at 4 °C. The anterior segment including the limbus was dissected in ice-cold PBS. The iris was carefully removed and shallow cuts were made to the anterior eye cup to display four quadrants in a petal-like pattern. The tissue was incubated in blocking solution of 0.3% PBST with 3% BSA for 2 hours at room temperature. Primary antibodies in blocking solution were added and the eyes left at 4°C for 2-3 days. After 4 PBST washes of an hour each at room temperature, secondary antibodies along with DAPI were added and eyes placed in the dark at 4 °C overnight. After 4 PBS washes of an hour each at room temperature, eyes were mounted on a slide, cover-slipped, and imaged.

For sections, B6. Prox1-GFP mice or B6 mice were cervically dislocated and eyes harvested with the optic nerve intact and fixed in ice-cold 4% PFA for 2 hours in 4 °C. A window was made to the back of the eye cup by removing the optic nerve. Eyes were equilibrated in 30% sucrose and embedded in OCT. Eyes were stored at -80°C until they were sectioned and processed for immunostaining. Sections were briefly washed in PBS and 0.3% PBST and blocked with blocking buffer made of 10% donkey serum in PBST. Primary antibodies were applied in blocking buffer overnight at 4°C and secondary antibodies with DAPI for 2 hours at room temperature. Slides were washed and cover-slipped.

## RNAscope *in-situ* hybridization

In-situ hybridization was carried out using RNAscope® Multiplex Fluorescent Reagent Kit v2- Mm (Advanced Cell Diagnostics a Bio Techne Brand, Newark, CA). Probes for mouse NPNT and SELP (also purchased from ACD Bio) were used. Fluorescent dyes Opal 690 and Opal 620 were used with each of these probes. Briefly, eyes from 3 month old C57BL/6J were enucleated and fixed in 4% PFA for 24 hr at 4°C. A window was made to the back of the eye cup, lens removed, and the anterior cup placed in 30% sucrose overnight at 4°C. Tissue was cryopreserved in OCT and placed at -80°C. Sections were cut at 12 µm thickness. RNAscope® was performed over two days as per manufacturer's protocol. Z stack images were taken using a confocal microscope at 40X magnification. A maximum projection image was then taken of the completed Z stack and edited using ImageJ.

## Confocal microscopy and postprocessing of images

Sections and whole mounts were imaged on a Leica SP8 confocal. Images were acquired at 40X resolution. 30-45µm Z-stack images of the tissue were acquired from whole mounts to image from the vascular plexus down to the trabecular meshwork cells. Images were analyzed and processed in Imaris.

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## Editors

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### Reviewer #1 (Public review):

Summary:

Balasubramanian et al. characterized the cell types comprising mouse Schlemm's canal (SC) using bulk and single cell RNA sequencing (scRNA-seq). The results identify expression patterns that delineate the SC inner and outer wall cells and two inner wall 'states'. Further analysis demonstrates expression patterns of glaucoma associated genes and receptor ligand pairs between SECs and neighboring trabecular meshwork.

Strengths:

While mouse SC has been profiled in previous scRNA-seq studies (van Zyl et al 2020, Thomson et al 2021), these data provide higher resolution of SC cell types, particularly endothelial cell (SEC) populations. SC is an important regulator of anterior chamber outflow and has important consequences for glaucoma.

Comments on the latest version:

The authors have addressed my primary concerns with the first version of the manuscript. This study represents a valuable resource in the molecular characterization of mouse Schlemm's canal cell types.

<https://doi.org/10.7554/eLife.96459.2.sa2>

### Reviewer #2 (Public review):

Summary:

This revised article has characterized the mouse Schlemm's canal expression profile using a comprehensive approach based on sorted SEC, LEC, and BEC total RNA-Seq, scRNA-Seq, and snRNA-Seq to enrich the selection of SECs. The revised study has successfully profiled genome-wide gene expression using sorted SECs, demonstrating that SECs have a closer similarity to LECs than BECs. The combined scRNA- and snRNA-Seq data with deep coverage of gene expression led to the successful identification of many novel biomarkers for inner wall SECs, outer wall SECs, collector channel ECs, and pericytes. In addition, the study also identified two novel states of inner wall SECs separated by new markers. The study provides significant novel information about the biology and expression profile of SECs in the inner and outer walls. It is of great significance to have this novel, convincing, and comprehensive study led by leading researchers published in this journal. The revision has improved the clarity and significance of the study with more details.

Strengths:



This is a comprehensive study using various data to support the expression characterization of mouse SECs. First, the study profiled genome-wide expression using sorted SECs, LECs, and BECs from the same tissue/organ to identify the similarities and differences among the three types of cells. Second, snRNA-Seq was applied to enrich the number of SECs from mouse ocular tissues significantly. Increased sampling of SECs and other cells led to more comprehensive coverage and characterization of cells, including pericytes. Third, the combined scRNA- and snRNA-Seq data analyses increase the power to further characterize the subtle differences within SECs, leading to identifying the expression markers of Inner and Outer wall SECs, collector channel ECs, and distal region cells. Fourth, the identified unique markers were validated for RNA and protein expression in mouse ocular tissues. Fifth, the study explored how the IOP- and glaucoma-associated genes are expressed in the scRNA- and snRNA-Seq data, providing potential connections of these GWAS genes with IOP and glaucoma. Sixth, the initial pathway and network analyses generated exciting hypotheses that could be tested in other independent studies.

#### Weaknesses:

The authors have addressed most of the previous comments by adding more details about the protocol and additional discussions. Several comments requiring additional experimental data have been addressed as future directions, such as protein validation, RNA expression validation in human samples, and GWAS-identified IOP genes.

#### Comments on the latest version:

The authors have addressed previous comments responsively. The authors have suggested several experiments to be completed in the future since these could be time-consuming with human samples. The revised article is with better clarity and clearer significance. No additional comments.

<https://doi.org/10.7554/eLife.96459.2.sa1>

#### Author response:

The following is the authors' response to the original reviews.

##### **Reviewer #1 (Public Review):**

###### *Summary:*

*Balasubramanian et al. characterized the cell types comprising mouse Schlemm's canal (SC) using bulk and single-cell RNA sequencing (scRNA-seq). The results identify expression patterns that delineate the SC inner and outer wall cells and two inner wall 'states'. Further analysis demonstrates expression patterns of glaucoma-associated genes and receptor-ligand pairs between SEC's and neighboring trabecular meshwork.*

###### *Strengths:*

*While mouse SC has been profiled in previous scRNA-seq studies (van Zyl et al 2020, Thomson et al 2021), these data provide higher resolution of SC cell types, particularly endothelial cell (SEC) populations. SC is an important regulator of anterior chamber outflow and has important consequences for glaucoma.*

We thank the reviewers for their thorough reading of our manuscript and their insightful comments.

#### Weaknesses:

*(1) Since SC has previously been characterized in mouse, human, and other species by scRNA-seq in other studies, this study would benefit from more direct comparisons to published datasets. For example, Table 4 could be expanded to list the SC cell numbers profiled in each study. Expression patterns highlighted in this study could be independently verified by plotting in publicly available mouse SC datasets. Further, a comparison to human expression patterns would assess whether type-specific expression patterns are conserved. Alternatively, an integrated analysis could be performed. Indeed, the authors mention that an integrated analysis was attempted but the data is not shown. It is unclear if this was because of a lack of agreement between datasets or other reasons.*

Table 4 now includes an expanded list of SC cell numbers in each study. We profiled the expression of *Npnt*, *Selp*, and *Ccl21a* in the Thomson et al., 2021 dataset and have included the concurring results in Figure S5. We were unable to do a similar profile using the Van Zyl, 2020 dataset due to small SC numbers. As previously mentioned, differences such as read depth, strain of animals used (including pigmented vs albino), method of cell isolation (including drug exposure), and number of cells profiled raise a significant impediment to integration with previously published datasets. A comparison to human atlas is a focus of future work.

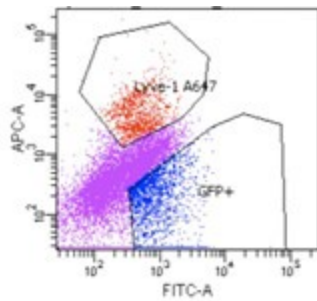
*(2) Figure 1 presents bulk RNA seq results comparing SEC, BEC, and LEC expression patterns. These populations were isolated using cell surface markers and enrichment by FACS. Since each EC population is derived from the same sample, the accuracy of this data hinges on the purity of enrichment. However, a reference is not given for this method and it is not clear how purity was validated. The authors later note that marker *Emcn*, which was used to identify BECs, is also expressed in SECs and LECs at lower levels. It should be demonstrated that these populations are clearly separated by flow cytometry.*

We have added the following clarifying text to the methods section: Forward and side scatter gates were first used to eliminate events with low scatter which include debris, cell fragments and pyknotic cells. Then propidium iodide positive dead cells were gated out. Further gating on the viable cells was applied such that distinct population of cells were isolated a) SECs: GFP+Lyve1-, b) LECs: GFP+ Lyve1+, c) GFP- BECs: Endomucin+.

We show here a representative of the flow sort showing the clear distinction in SEC and LEC cell isolation.

#### Author response image 1.

Flow sorted SEC and LEC. We obtained two distinct populations; 1. SEC cells (GPF+LYVE1-- blue) 2. LEC (GPF+LYVE1+- red). Note eFluor 660 emission was collected using the Alexa647 (A647) setting of the flow cytometer. Additionally, SEC marker expression from bulk RNA-seq aligns with signature gene expression from SECs in single cell RNA-seq (Figure S3).



(3) Bulk RNA-seq analysis infers similarity from the number of DEGs between samples, however, this is not a robust indicator. A correlation analysis should be run to verify conclusions.

We have provided a heatmap with hierarchical clustering based on Euclidean distance of the EC subtypes (Figure 1B) analyzed by bulk RNA seq in addition to number of DE genes between subtypes.

(4) Figures 2-4 present three different datasets targeting the same tissue: 1) C57bl/6j scRNA-seq, 2) C57bl/6j snRNA-seq, 3) 129/sj scRNA-seq. Integrated analysis comparing datasets #1 to #2 and #3 is also presented. Integration methods are not described beyond 'normalization for cell numbers'. It is unclear if additional alignment methods were used. Integration across each of these datasets needs careful consideration, especially since different filtering methods were used (e.g. <20% mito in scRNA-seq and <5% in snRNA-seq). Improper integration could affect the ability to cluster or exaggerate differences between cell/types and states. It would be useful to demonstrate the contribution of different samples and datasets to each cell type/state to verify that these are not driven by batch effects, mouse strain, or collection platform.

We agree that integration should be performed with careful consideration to confounding factors. We demonstrate the contribution of different samples and datasets to show how our datasets integrated well (we had added panels to Figure 3C and 4C) and that cell types/states contribution was uniformly distributed across methods (C57BL/6J single cell and single nuc) and backgrounds (C57BL/6J and 129/Sj) were not a result of integration.

(5) IW1 and IW2 are not well separated, and it is unclear if these represent truly different cell states. Figure 5b shows the staining of CCL21A and describes expression in the 'posterior portion' but in the image there are no DAPI+ nuclei in the anterior portion, suggesting the sampling in this section is different from Figure 5a. This would be improved by co-staining NPNT and CCL21A to demonstrate specificity.

Since both our antibodies are derived from the same species (goat), a co-labeling wasn't possible. To be prudent, we used adjacent sections, flat-mounts, and RNAscope and provided further evidence of the anterior/posterior "bias" in supplemental figures.

(6) The substructures observed within clusters in sc/snRNA-seq data suggest that overall profiling may still not be comprehensive. This should be noted in the discussion.

We agree and have added this note in the discussion: "With greater sampling and deeper transcriptomic depth, it is likely that additional SEC cell states/types will be identified."

## Reviewer #2 (Public Review):

### Summary:

*This article has characterized the mouse Schlemm's canal expression profile using a comprehensive approach based on sorted SEC, LEC, and BEC total RNA-Seq, scRNA-Seq, and snRNA-Seq to enrich the selection of SECs. The study has successfully profiled genome-wide gene expression using sorted SECs, demonstrating that SECs have a closer similarity to LECs than BECs. The combined scRNA- and snRNA-Seq data with deep coverage of gene expression led to the successful identification of many novel biomarkers for inner wall SECs, outer wall SECs, collector channel ECs, and pericytes. In addition, the study also identified two novel states of inner wall SECs separated by new markers. The study provides significant novel information about the biology and expression profile of SECs in the inner and outer walls. It is of great significance to have this novel, convincing, and comprehensive study led by leading researchers published in this journal.*

### Strengths:

*This is a comprehensive study using various data to support the expression characterization of mouse SECs. First, the study profiled genome-wide expression using sorted SECs, LECs, and BECs from the same tissue/organ to identify the similarities and differences among the three types of cells. Second, snRNA-Seq was applied to enrich the number of SECs from mouse ocular tissues significantly. Increased sampling of SECs and other cells led to more comprehensive coverage and characterization of cells, including pericytes. Third, the combined scRNA- and snRNA-Seq data analyses increase the power to further characterize the subtle differences within SECs, leading to identifying the expression markers of Inner and Outer wall SECs, collector channel ECs, and distal region cells. Fourth, the identified unique markers were validated for RNA and protein expression in mouse ocular tissues. Fifth, the study explored how the IOP- and glaucoma-associated genes are expressed in the scRNA- and snRNA-Seq data, providing potential connections of these GWAS genes with IOP and glaucoma. Sixth, the initial pathway and network analyses generated exciting hypotheses that could be tested in other independent studies.*

We thank the reviewer for their comments on the strengths of this study.

### Weaknesses:

*A few minor weaknesses have been noted. First, since snRNA-Seq and scRNA-Seq generated different coverage of expressed genes in the cells, how did the combined analyses balance the un-equal sequencing coverage and missing data points in the snRNA-Seq data? Second, the RNA/protein validation of selected SEC molecular markers was done using mouse anterior segment tissues. It would be more helpful to examine whether these molecular markers for SECs could work well in human SECs. Third, the effort to characterize the GWAS-identified IOP- and glaucoma-associated genes is exciting but with limited new information. Additional work could be performed to prioritize these genes.*

Integration of sc-Seq and sn-Seq data: We have addressed a similar integration question from reviewer 1 and have now included a plot showing the distribution of cells upon integration. Integration methods are not perfect and generally result in some loss of data especially when datasets of un-equal sequencing coverage are integrated. However, we did not observe any obvious differences between the original (un-integrated) and integrated datasets. We also noted that cell types/states contribution was similarly distributed across methods (C57BL/6J

single cell and single nuc) and backgrounds (C57BL/6J and 129/Sj) and clustering were not a result of batch-effects.

We agree about the human relevance of SEC markers, and this will be a focus of future work.

Another focus of our future work is to understand how GWAS identified IOP and glaucoma genes change in disease states.

**Recommendations for the authors:**

**Reviewer #1 (Recommendations For The Authors):**

*Minor:*

*(1) Figure 5- DAPI should be listed in the legend.*

*(2) Figure 5- It would be helpful to label IW1 and IW2 regions in the UMAPs.*

We have incorporated the suggestions in Figure 5 and legend.

**Reviewer #2 (Recommendations For The Authors):**

*(1) The study has validated RNA/protein expression of the selected biomarkers for IW/OW SECs in mouse eyes. It would be more helpful to confirm that these newly identified molecular biomarkers for SECs could apply to human eyes. This could be examined through available human scRNA-/snRNA-Seq data or targeted RNA and protein staining experiments. The additional validation in human SECs would make the current discovery more convincing.*

We agree with the importance of validation in human samples, and is the scope of future work.

*(2) The combination of scRNA-Seq and snRNA-Seq from three batches of experiments increased the statistical power to identify subtypes of SECs. It would be helpful to include more details on how the qc, missing data, and normalization across different batches were dealt with.*

We have incorporated more details in the methods section of the paper.

*(3) The authors explored the underlying molecular connection between the newly identified IOP/glaucoma-associated genes using the newly generated SEC-targeted scRNA/snRNA-Seq data. Many of these associated genes were present in the same SEC cells. It would be interesting to see how many of these genes' expression levels are correlated with each other via a network. These potential correlation networks across SECs could lead to identifying novel upstream regulators or network hubs, which could target many IOP-associated genes for future studies.*

We agree with the importance of a correlation network analysis, but this is a focus of future work, especially in normal and disease states.

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